



Li-ion batteries using electrolytes based on mixtures of poly(vinyl alcohol) and lithium bis(trifluoromethane) sulfonamide salt



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ABSTRACT

Poly(vinyl alcohol) (PVA) has, to a limited degree, previously been investigated as an alternative host material to the dominating poly(ethylene oxide)-(PEO)-based polymer electrolytes used in electrochemical applications. These PVA-materials were found to be highly ionically conducting for a number of different salts at temperatures below the glass transition temperature, but in the presence of DMSO (dimethyl sulfoxide) residues. Here, we investigate the PVA:LiTFSI (lithium bis(trifluoromethane) sulfonamide) electrolytes system, thereby for the first time using the salt most commonly explored in PEO-based systems also in PVA. It is shown that this results in significantly higher conductivities than for alternative Li salts. Comparisons between electrolytes casted in DMSO, H₂O and hot-pressed samples confirm the necessity of solvent residues present in the DMSO casted films for the ionic transport, although no obvious liquid phase is formed. Moreover, we also show the feasibility of PVA-based electrolytes by construction of functional Li-metal | PVA-LiTFSI (DMSO) | LiFePO₄ batteries operating at 60 °C, displaying stable capacities of 136 mAh/g LiFePO₄ at a rate of C/20.

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1. Introduction

Because of their high energy densities, lithium-ion batteries are promising solutions to the energy storage needed to accommodate fluctuations in the electrical grid during peak hours when changing to renewable energy sources [1,2]. However, large scale applications lead to high risks in safety mainly due to the liquid organic compounds used in the electrolyte. To improve safety, one strategy would be to replace – or partially replace, as in gel polymer electrolytes – the liquids with a solid polymer which show superior properties in terms of improved electrochemical stability, reduced risk of leakage and fire hazards [3,4].

A major challenge for the conventional poly(ethylene oxide) (PEO) electrolytes is their low ionic conductivity at room temperature. This is because in PEO:salt systems, ions move as a result of segmental motion of the polymer backbone in the amorphous phase. Hence, low glass transition temperatures and reduced crystallinity are desirable in order to maximize ionic conductivity, which has led the research community to primarily investigate amorphous polymer salt systems [5]. In recent years, research efforts have focused on exploring and revisiting

alternative polymer materials for their use in polymer electrolytes. Polycarbonate/polyester electrolytes, for instance, may allow near room temperature battery operation [6].

One strategy to increase the conductivity, even below T_g, would be to generate materials with the ionic conduction decoupled from the segmental motion, thereby shifting the conduction profile from VTF to an Arrhenius behavior. Rigid polymers with difficulties in close packing have been studied in this context [7]. Another alternative are hydrogen bonding polymers such as poly(vinyl alcohol) (PVA), which has showed good ability to dissolve alkali-metal salts and high ionic conductivity [8–16]. However, it has been argued that this is due to the presence of DMSO solvent residues, and solvent-free fabrication methods have displayed very limited conductivity [16]. In this study, PVA-based electrolytes using LiTFSI (lithium bis(trifluoromethane)sulfonamide), a salt with well-known plasticizing properties are explored for the first time, and compared with earlier studied salts (LiTf and LiClO₄). We also investigate the effect of solvent residuals by using H₂O, which can easily be evaporated, as alternative solvent during casting, as well as hot-pressing techniques, in order to achieve a better understanding of the ion conduction mechanism in the materials. Moreover, we construct and bench-mark the first Li-battery using a PVA-based electrolyte.

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2. Experimental

2.1. Electrolyte and electrode preparation

All chemicals were purchased from Sigma Aldrich and used as received unless other is specified. LiTFSI, LiCF_3SO_3 and LiClO_4 were dried at 120°C overnight under reduced pressure in an Ar-filled glovebox prior to use. Free standing films were produced via solution casting of the dissolved (by heating) polymer (Mowiol 20-98, $M_w \sim 125,000$) and salt in dedicated PTFE moulds. The solvent was removed over the course of 5 days in a vacuum furnace placed in an Ar-filled glovebox. Hot-pressed films were made by first ball-milling polymer and salt mixtures for 1.5 h at 300 rpm using a planetary ball mill. Films were then hot-pressed with a force of 3 tons at 200°C . Water casted films were produced via the same method as using DMSO, although outside of a glovebox.

2.2. Characterization

Electrochemical Impedance Spectroscopy (EIS) was performed by sandwiching electrolyte films between two stainless steel blocking electrodes in an air-tight Swagelok cell setup. Data was collected on the cooling run between 115 and 25°C using a Schlumberger SI 1260 Impedance/Gain-phase analyzer between a frequency interval of 1 Hz – 10 MHz and an amplitude of 10 mV . The ionic conductivity was determined from the width of X-axis intercept of the semi-circle in the Nyquist diagrams. Fourier Transform Infrared (FTIR) Spectroscopy was conducted using a Perkin Elmer Instruments Spectrum One equipped with a ZnSe ATR sample holder between 4000 and 650 cm^{-1} with a resolution of 4 cm^{-1} for 16 scans. For Differential Scanning Calorimetry (DSC), samples were sealed in air-tight aluminum pans inside of a glovebox and loaded onto a TA-instrument Q200. The pans were put on a heat-cool-heat cycle from room temperature to 150°C , cooled to -20°C and then heated to 150°C at $5^\circ\text{C}/\text{min}$. T_g was determined from the DSC data as the inflection point of the transition slope. Residual DMSO content was analyzed by Thermogravimetric Analysis (TGA) using a TA-instruments TGA Q500. Samples were heated at $5^\circ\text{C}/\text{min}$ using N_2 as protective gas.

2.3. Battery testing

Cathodes of LiFePO_4 were made by solution casting a slurry consisting of 80 wt% LiFePO_4 , 10 wt% carbon black and 10 wt% of PVA binder on top of carbon coated Al-foil current collector. After punching circular discs, cathodes were dried at 120°C overnight under reduced pressure in an Ar-filled glovebox. The electrolyte (LiTFSI/PVA, molar ratio 4:1 OH:Li) was then casted on top of the cathode from solution via the same procedure as described above. Li-metal foil (Foote Minerals, $125\text{ }\mu\text{m}$ thick) was used as received.

Cells were then assembled in pouch cell format using polymer coated Al-foil.

The cells were cycled galvanostatically at 60°C on an Arbin battery tester at a constant current density of $15.4\text{ }\mu\text{A cm}^{-2}$ (corresponding to a cycling rate of C/20) between the cut-off potentials of 2.8 and 3.6 V, apart from the first cycle which had a cut-off at 4.0 V. The first lithiation of LiFePO_4 took 35 h, as a result of irreversible reactions above the insertion potential. In order to minimize these reactions, the step time was limited to 24 h.

3. Results and discussion

As mentioned in the introduction, PVA-based electrolytes have been reported to possess high ionic conductivity, but it has also been shown by MacFarlane and co-workers that this has been achieved only when in presence of DMSO residues in the films [16]. It is therefore of interest to investigate if DMSO residues are the main cause of the conductivity by making films with other solvents such as water or solvent free hot-pressing, and a variety of salts. The amount of salt was chosen at a molar ratio 4:1 (OH:Li), which displayed the highest conductivity in the case of PVA- LiCF_3SO_3 when investigated by MacFarlane et al. [16]. In addition, since LiTFSI has not been tested in the PVA-DMSO system previously, a variety of salt contents were tested ranging from molar ratios of 100:1 up to 4:1 (OH:Li). No traces of undissolved LiTFSI could be observed using X-ray diffraction, also for the highest concentrations (see Supplementary Information). Higher salt contents led to very high viscosities, making casting of the solution difficult.

3.1. Thermogravimetric Analysis

The amount of residual DMSO was investigated by TGA, and it was found that at least 10 wt% of DMSO was indeed present in the films after rigorous drying as can be seen in Fig. 1. Replacing DMSO as solvent with water showed very small ($<1\%$) changes in mass for the LiTFSI-based samples, indicating that most of the solvent had evaporated during the drying procedure. Our levels of residual DMSO (ca 10%) are comparable with previous studies; Yamamoto et al. conclude that 3.5% DMSO remains in a LiCF_3SO_3 /PVA film after vacuum drying at elevated temperature using a GC method [8] whereas MacFarlane show 12% residual DMSO for a 50 wt% of LiCF_3SO_3 after similar treatment [16]. Yamamoto also concluded that DMSO evaporated completely from DMSO/ LiCF_3SO_3 mixtures when treated in vacuum pointing towards a solvent interaction with PVA rather than the salt [8]. On the other hand, there seem to be a correlation between DMSO residues and salt concentration in LiCF_3SO_3 /(polyacrylonitrile) PAN-based solid polymer electrolytes casted from DMSO solutions [19]. However, the residual levels are much higher (ca 50%) indicating property differences between PAN (nitrile) and PVA (hydroxyl).

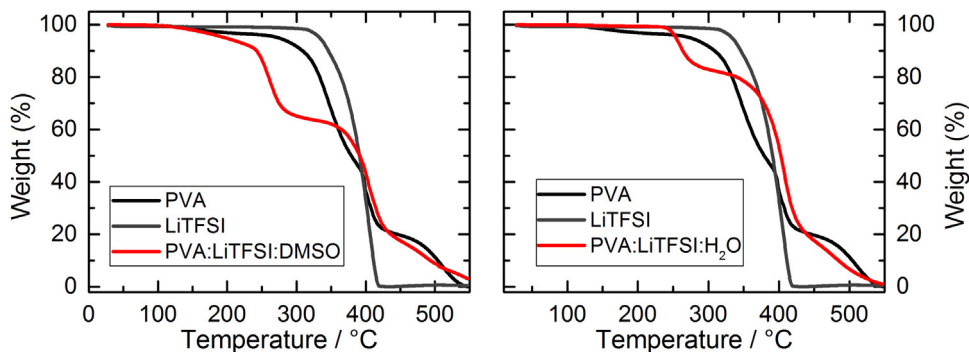


Fig. 1. Thermal decomposition of PVA-LiTFSI electrolyte (molar ratio 4:1 OH:Li) containing DMSO (left) or water (right) residues.

The impact of having solvent residues (10% DMSO) or only tiny amounts ($<1\%$ H_2O) on the ionic conductivity was investigated by impedance spectroscopy.

3.2. Electrochemical Impedance Spectroscopy

Presented in Fig. 2a is the ionic conductivity measured for increasing salt content in the PVA-LiTFSI-DMSO system. With increasing salt concentration, the number of charge carriers increases, reaching a peak value in ionic conductivity at 62 wt% salt (4:1 OH:Li molar ratio), which was the maximum achievable amount.

The temperature dependence of these PVA-LiTFSI-DMSO systems is presented in Fig. 2b. The ionic conductivity is seen to be significantly higher for the higher salt concentrations, which is believed to be due to the larger amount of charge carriers present. The slope of the curves also shows less temperature dependence at high salt concentrations which indicates that ionic motion is not

only assisted by polymeric motion. Electrolytes with a 4:1 or 10:1 OH:salt ratio display a constant slope in the Arrhenius plot in the temperature range investigated, rather than the Vogel-Tammann-Fulcher (VTF) behavior associated with polymer segmental motion. Low temperature data points for the 100:1 OH:salt ratio were found to be virtually non-conducting ($<10^{-12} \text{ S/cm}$).

The impact of residual DMSO left in the electrolyte films is seen when comparing results given in Fig. 2c and 2d, where conductivities are shown for films casted using both DMSO (2c) and water (2d) with several different salts. When changing the solvent to H_2O , a clear drop in ionic conductivity by about one order of magnitude is observed, which is similar to earlier observations when comparing DMSO films with hot-pressed samples [16]. The water casted samples in Fig. 2d all show constant slopes in the Arrhenius plot above 40°C indicating that ion conduction is very loosely linked to polymer backbone motion.

For all investigated systems, LiTFSI-based electrolytes display higher conductivities, which might be due to the plasticizing effect

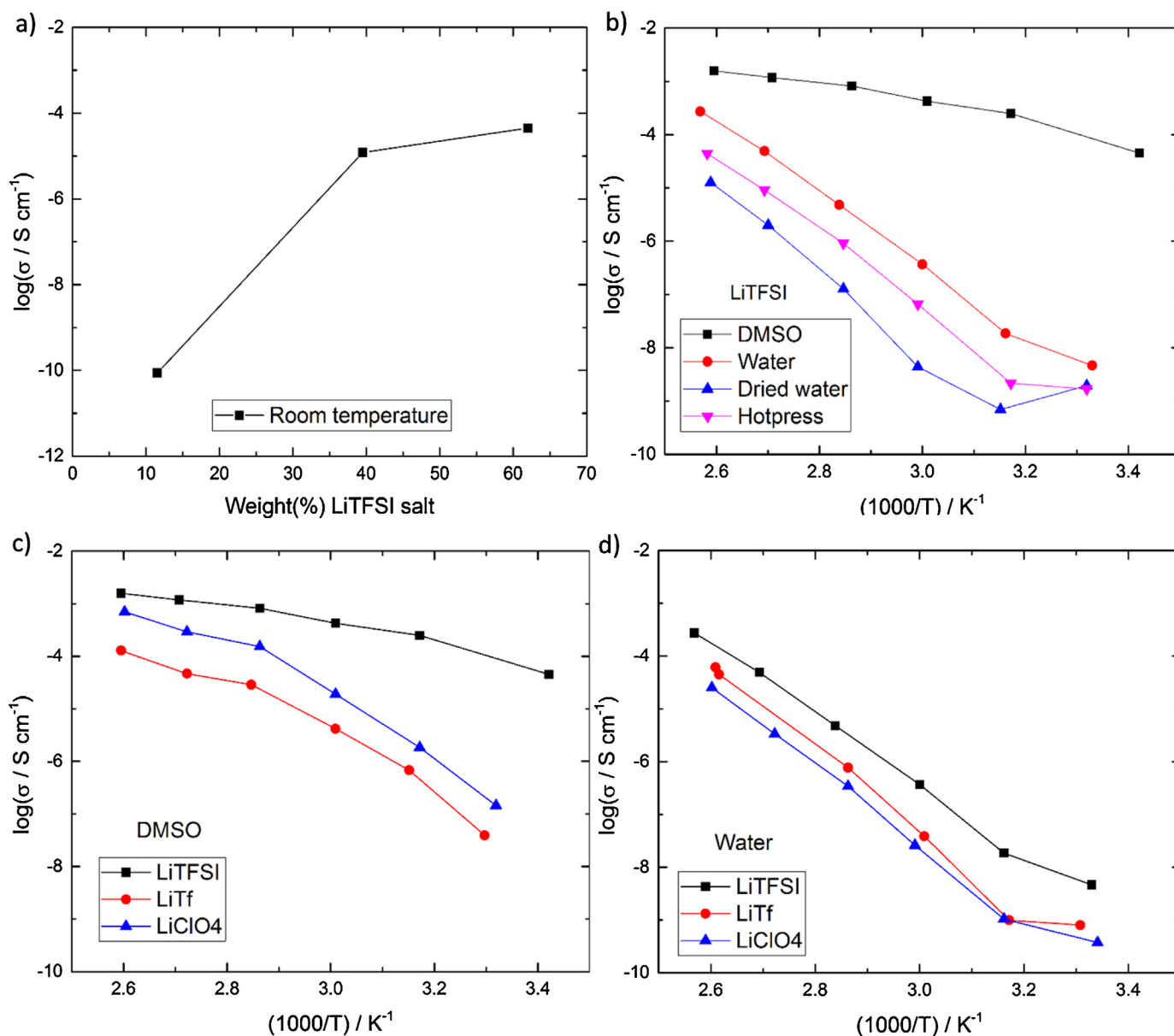


Fig. 2. Ionic conductivity for different PVA-based electrolytes: a) increasing concentration of LiTFSI (casted using DMSO); b) Temperature dependence of the DMSO-casted LiTFSI samples; c) Electrolytes casted with DMSO as solvent using different salts (molar ratio 4:1 OH:Li); d) Electrolytes casted using water instead of DMSO (molar ratio 4:1 OH:Li).

of the anion or the higher mobility of the anion itself. The latter might seem more plausible considering the Arrhenius-like behaviour of the conductivity, but a plasticizing anion could also contribute a better percolating network for the solvent residues. The TFSI anions are also likely to have less interaction with the PVA-backbone than the alternatives, which are likely to form relatively strong bonds to the OH-groups. TFSI has part of its negative charge centered on the central nitrogen atom, while the ClO_4^- and CF_3SO_3^- have more localized electrons on the oxygen atoms. It is also interesting to note that the depression in conductivity is less significant for the LiCF_3SO_3 salt. This might be due to the hygroscopic nature of this salt, which makes the electrolyte retain a larger fraction of solvent after drying.

3.3. Fourier transform Infrared Spectroscopy

The IR-spectra of the casted samples (blue and red in Fig. 3) all have slightly shifted salt peaks in comparison to the powder signals (green), which originates from the samples meshed together [17]. The CF_3 anti-symmetric stretch at 1327 cm^{-1} is shifted towards 1325 cm^{-1} , the SO_2 stretch at 1201 cm^{-1} to 1183 cm^{-1} and the S_2N peaks at 811 cm^{-1} and 746 cm^{-1} show peak shift to 791 cm^{-1} and 739 cm^{-1} , respectively. To be noted is that the shifts are very similar when casting with water and DMSO respectively, with the casted (red) line. The similarities indicate that the residual DMSO does not interact strongly with the TFSI

anion of the salt, but instead influences the characteristic OH-stretching around 3500 cm^{-1} of the casted films, causing a shift along with a shape distortion compared to both mixed powders (green) and pure PVA (grey). This interaction between DMSO and functional groups could influence the nature of the hydrogen bonds, enabling percolation pathways for faster ion transport. However, interactions between the ionic species, the solvent residuals, the -OH groups or adducts thereof cannot be fully ruled out. The appearance of the sulfoxide peak is further evidence of the presence of solvent left in these electrolytes.

3.4. Differential Scanning Calorimetry

The hot-pressed samples, which were prepared entirely solvent free show very similar glass transition temperatures as the water casted samples; see Table 1. This is further indication of the drying procedure removing most of the water used during casting as discussed in section 3.1. A clear difference is seen instead for the triflate salt where a drop in T_g can be observed when using water as the solvent. This could, again, be due to the triflate salt adsorbing residual water. Films with LiClO_4 have the lowest glass transition temperatures, but not the highest ionic conductivity, indicating that the ionic conductivity is not strongly coupled to polymer segmental motion. This can be seen from the water-casted electrolytes, where the different T_g values (Table 1) are poorly correlated with the shape of the VTF-curves in Fig. 2d.

DSC analysis of glass transition temperatures showed no clear transitions for any of the DMSO casted electrolyte films, which could be due to the films having transitions outside of the measured temperature range (i.e. below -20°C). T_m for DMSO is 19°C , but no transition exo- or endotherm was observed around this temperature, or below for possible DMSO salt phases, showing that within the measured range (-20°C – 150°C) no true liquid phase exists in the system.

3.5. Galvanostatic cycling

DMSO- and water-casted PVA:LiTFSI electrolyte (4:1 OH:Li) were tested in Li:LiFePO₄ battery setups at 60°C . Cells with water-casted PVA-electrolytes, however, failed to pass current even at cycling rates of C/50 or less, clearly demonstrating their insulating nature. An instant potential jump was noted after a current was applied, suggesting that the electrolyte did not allow for any significant lithium-ion conduction.

The cell with DMSO-casted PVA-electrolytes on the other hand, could be operated at slow cycling rates of C/20. As mentioned above, this electrolyte contained residual DMSO, which may allow lithium-ion diffusion in the polymer gel. The discharge capacities and corresponding coulombic efficiencies over the first 20 cycles are presented in Fig. 4. A low Coulombic efficiency could be noted on the first cycle, probably due to additional irreversible reactions occurring just above the redox potential of the cathode. Therefore, the upper cut-off potential could not be reached. In order to bypass this issue, a time limitation was used on following cycles that ended the charge/discharge reaction before the cell reaction

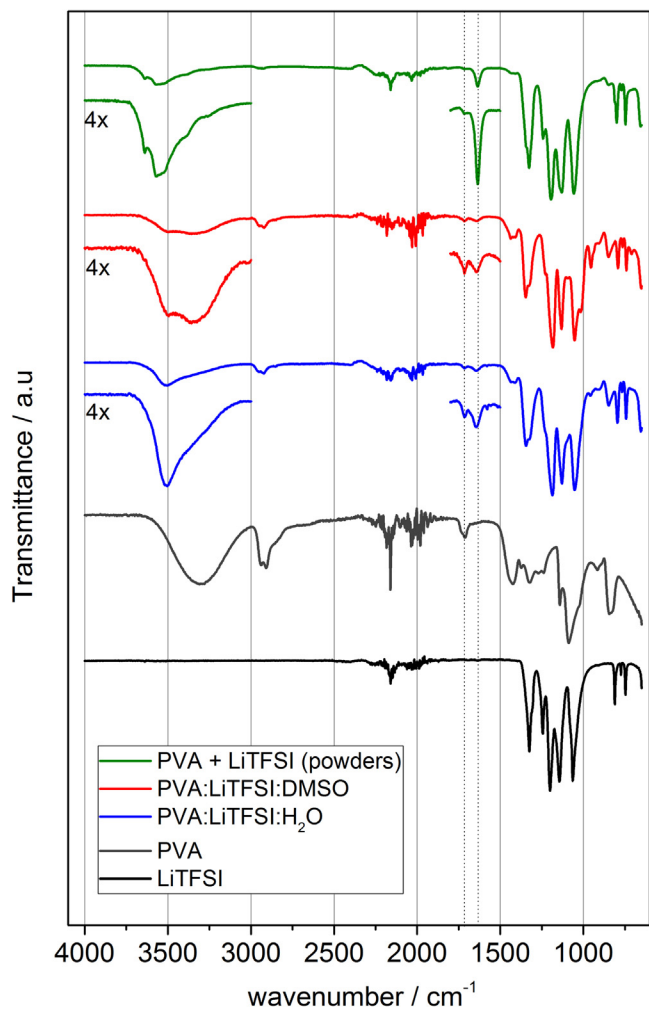


Fig. 3. FTIR spectrum of PVA-LiTFSI samples (molar ratio 4:1 OH:Li) taken in ATR mode.

Table 1

Glass transition temperatures for different PVA-based electrolyte films.

Solvent	Salt	$T_g/^\circ\text{C}$
Water	LiTFSI	58
Water	LiTF	67
Water	LiClO_4	41
Hot-pressed	LiTFSI	59
Hot-pressed	LiTF	82
Hot-pressed	LiClO_4	40

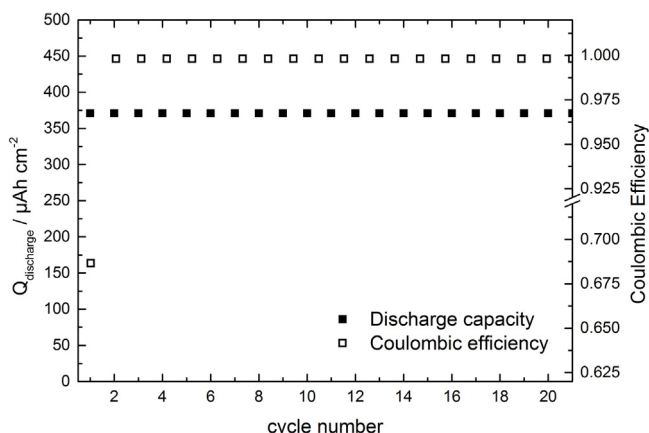


Fig. 4. Discharge capacity and Coulombic efficiency for an electrochemical cell comprising Li-metal | PVA-LiTFSI (DMSO) | LiFePO₄ cycled galvanostatically at C/20 and 60 °C. The capacity is equivalent to ca. 136 mAh/g LiFePO₄.

finished. The capacity limitation then allowed stable cycling over 20 cycles and Coulombic efficiencies close to unity (99.81%).

It was reported previously by Yamamoto [11] that PVA-LiCF₃SO₃ membranes casted from DMSO show reactions with lithium at 80 °C. However, it was also noted that irreversible reactions, for instance from reactions of the OH functional groups of PVA with lithium, were not observed during operation of a double layer capacitor. In our study, an experiment using PVA-LiCF₃SO₃ as an electrolyte (in contrast to the LiTFSI congener) was conducted but failed within 4 cycles. The PVA-LiTFSI electrolyte thus appears to operate more stably in the given battery chemistry. In a related context, it has been shown that methanol:Lil form a solid adduct with a 4:1 stoichiometry between OH-groups and iodide and that the ionic conductivity is rather high ($\sigma = 2.7$ mS/cm) at room temperature, but even more strikingly show a stability towards metallic lithium despite the presence of hydroxyl groups [18].

In the case of the presented PVA-LiTFSI electrolyte herein, a continuous decrease in the lower cut-off potential could also be noted (see the supplementary information). This drop becomes more pronounced towards the end of the 20 cycles, which can be interpreted as a degradation process, although it is unclear if it originates from either of the electrodes or residual DMSO.

4. Conclusions

It was found that PVA can dissolve up to about 60 wt% of LiTFSI salt and reach conductivities as high as 4.5×10^{-4} S/cm at 60 °C and 4.5×10^{-5} S/cm at room temperature in the presence of DMSO residues. When replacing the solvent DMSO with water, the ionic conductivity dropped about one order of magnitude due to better

elimination of solvent residues as analyzed by TGA. In contrast to PVA-electrolytes employing LiCF₃SO₃ and LiClO₄ respectively, PVA-LiTFSI electrolytes show very little temperature dependence and nearly linear conductivity behavior in the Arrhenius plots. FTIR analysis indicated evidence of interactions between the OH functional group of the polymer and the salt. Cycling data of Li-metal | PVA-LiTFSI (DMSO) | LiFePO₄ showed stable performance for at least 20 cycles. However unwanted side reactions take place which have a negative impact on cell performance. Cells using electrolytes without DMSO showed too poor device performance for evaluation, again displaying the importance of solvent present in PVA-based electrolyte systems.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.electacta.2017.05.127>.

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